# Data bibliography / attribution

## Orbital Technosphere / Global

### 01 CelesTrak / orbital elements and memberships

26 OMM CSV catalogs. Collected 2026-11-03T17:13:12Z to 2026-11-03T17:14:05Z.
Active catalog supplies the satellite population; membership catalogs identify named fleets. Three bounded debris catalogs supplement it.
Catalogs: active, starlink, oneweb, kuiper, qianfan, hulianwang, gnss, iridium-next, globalstar, orbcomm, resource, science, stations, fengyun-1c-debris, iridium-33-debris, cosmos-2251-debris, planet, spire, intelsat, ses, eutelsat, telesat, gps-ops, glo-ops, galileo, beidou.
Attribution: CelesTrak. Exact feed URLs, element epochs and hashes are retained in the source ledger.
https://celestrak.org/NORAD/documentation/gp-data-formats.php

### 02 SGP4 / Vallado, Crawford, Hujsak and Kelso

Positions propagated from the pinned elements using the reference SGP4 implementation, WGS-72 and AFSPC mode.
Calculation instant: 2026-11-03T17:10:00Z. 44 records excluded by epoch; 0 propagation failures.
Public element uncertainty and approximate Earth rotation limit these visualization positions.
https://celestrak.org/publications/AIAA/2006-6753/

### 03 Natural Earth / terrestrial context

Natural Earth 1:10m physical land provides the subdued geographic background.
Attribution: Natural Earth. Public-domain cartographic data.
https://www.naturalearthdata.com/downloads/10m-physical-vectors/10m-land/

### Historical validation / NASA SSCWeb

Reference instant: 2026-08-13T15:36:00Z.
Historical regression reference only. The November refresh was unreachable; this response does not provide an independent comparison of the plotted November positions.
https://sscweb.gsfc.nasa.gov/WebServices/REST/

### Cartography / Cartofreako

Projection: Authagraph. The projection bibliography credits its mathematical and design sources.
Separate map and legend composition follows the Setouchi ST2 layout. Typography: Atkinson Hyperlegible; bundled fonts under the SIL Open Font License.
https://bdekoz.github.io/cartofreako/docs/pages/projections/authagraph/bibliography.html

### Pinned inputs / source ledger

The profile records the calculation time, observer, filters and every catalog URL. Adjacent acquisition.json and SHA256SUMS record retrievals and original input bytes.
Map and legend carry the same timestamp and counts; the complete acquisition ledger is embedded in this SVG.
Map image scaled to 98.863636% with proportional sheet margins.
https://github.com/bdekoz/cartofreako/blob/main/assets.static/orbital-technosphere/20261103/orbital-technosphere-profile.json

Setouchi AuthaGraph: +4.35-inch printed horizontal move on a 43.5-inch image. Antarctic land is deliberately displaced to the lower left; object coordinates retain source registration.

### MSP speculation 4 / palette 2 / Ultramarine / porcelain

Randoma11y: ground #193fb8; type #f5eae7. sRGB contrast 7.28:1 (screen only). Vendor markers retain source colors.
https://randoma11y.com/%23193fb8/%23f5eae7
Epson Stylus Pro 9900 / metallic glossy paper / cyan and light magenta reported low; grays reportedly print violet.
Intended RGB hues; final printed hue is unmeasured. No green correction, ink separation or metallic reflection simulation.

## SGP4 positions / 2026-11-03T17:10:00Z

17,909 plotted objects / named fleets and residual roles

*Seven-day element age; six-hour future tolerance.*

- Starlink 11155
- OneWeb 651
- Kuiper 391
- Qianfan 256
- Hulianwang 235
- Iridium 80
- Globalstar 36
- ORBCOMM 14
- Planet 129
- Spire 72
- Intelsat 55
- SES 67
- Eutelsat 29
- Telesat 17
- GPS 32
- GLONASS 24
- Galileo 31
- BeiDou 52
- Debris 1101
- Other active 3247
- Other navigation 19
- Other Earth observation 154
- Science 43
- Human presence 19

One object per NORAD ID. Overlapping named memberships use profile order.

Starlink NORAD IDs are visible at 6 pt in a separate id layer. All name layers are hidden.

Black x: debris. Sunburst: Starlink. Square: Intelsat. Triangle: remaining Earth observation.

Active-marker opacity records illumination. Debris remains solid black.

Terrestrial subpoints; bounded representative ground tracks; reference site marked.

Starlink sunbursts: 1.5× linear size on both orbital maps.

Starlink text: 6 pt; all satellite markers paint above it.

All terrestrial land, including Antarctica, paints below the data.

# Orbital technosphere
# Global / Authagraph

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